

MECHDIAG AI

Machinery Condition Monitoring Diagnostic Agent

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The Problem

Diagnosing mechanical faults relies on scarce, specialized expertise. The current predictive maintenance paradigm faces critical bottlenecks:

- **Interpretation Complexity:** Junior technicians struggle to cross-reference FFT spectra, phase readings, and defect frequencies manually.
- **Data Fragmentation:** Engineers must manually correlate sensor data with dense, 500-page PDF service manuals.
- **Economic Impact:** Misdiagnosed faults lead to premature part replacement, unexpected catastrophic failure, and millions in downtime.

The Solution: MechDiag AI

A virtual Senior Vibration Analyst powered by a Large Language Model agent capable of **strict diagnostic reasoning**.

Multi-Modal Ingestion

Instantly parses raw vibration inputs alongside unstructured equipment manuals and thermal data.

Agentic Reasoning

Routes data through logic trees (ISO 13374, ISO 20816) and identifies missing measurements instead of guessing.

Deployment & Validation

Unlike traditional dashboards that passively display data, MechDiag AI actively reasons through the logic tree to prevent operational downtime.

Fully Interactive Web Application

<https://mechdiag-ai-gc4gzdgxqmbyszjenykqvre.streamlit.app/>

"Democratizing expert-level predictive maintenance to save global industries **millions in downtime**."

